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From Ideas to Implementations: Closing the Gaps between Technical Experts and Software Solutions

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ABSTRACT
Rapid delivery strategies strive to balance critical performance qualities vs. reducing the time between an idea and deployment of a software implementation of that idea. For industrial software solutions that encapsulate expertise in deliverable components, technical SMEs (Subject Matter Experts) with ideas and knowledge have traditionally partnered as requirements providers with software development teams. These human processes are not optimally fast, are vulnerable to errors in translating or interpreting requirements, and do not scale when software teams need to integrate the knowledge of many SMEs into multiple software solutions and deployments. To address these limitations, ABB has pursued an industrial research initiative for innovative SME toolsets with focus on two goals: to accelerate the creation, evolution, reuse, and delivery of expert algorithms, and to streamline the deployment of these algorithms into releases and fielded solutions. The vision underpinning the initiative is to empower technical SMEs as "end-user developers" to convert their knowledge into reusable software solution components without having to learn, perform, or partner on traditional software development, integration, or deployment. In this paper, we summarize our experiences and lessons learned to date from this initiative, key continuing challenges, and some positional thoughts on how end-user development by technical SMEs aligns with emerging approaches for rapid delivery and evolution.

CCS Concepts
• **Social and professional topics~Socio-technical systems** • *Applied computing~Business intelligence* • *Software and its engineering~Application specific development environments.*

Keywords
Subject Matter Expert; Industrial Analytics; End-User Programming; End-User Software Engineering; Visual Programming Environments; Continuous Deployment

1. INTRODUCTION
In industrial software, achieving greater agility is distinctly different from other domains. First, long development lifecycles and product lifetimes are still common in modern industrial enterprises [8]. Industrial software application ecosystems thus need careful architecture to enable them to evolve over years (or decades) to handle fundamental changes in operating systems and technologies. Second, for critical operational technology software, security and validation concerns are paramount. Frequent releases can thus be undesirable to some industrial or military customers. The same concerns have driven reluctance to adopt provider-hosted platforms and cloud-based services where costs and risks of updates can be managed with fewer operational disruptions.

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However, two emerging forces are mitigating these factors. As interest grows in adoption of Industrial Internet of Things (IIoT), customers are warming to migration of some of their industrial applications from closely-managed on-premise installations to provider-hosted solutions. In parallel, operational software systems increasingly leverage architecture approaches that accommodate carefully-controlled extensions and customizations. This maximizes the continuing value of well-established and validated systems, with less overhead or risk vs. major updates.

Accordingly, in striving for greater agility and speed to market in serving industrial customers, IIoT solution providers like ABB (www.abb.com) increasingly pursue faster or even continuous deployment [3] of new capabilities, and greater customizability and extensibility of fielded solutions. Our industrial research initiative is exploring how to enable end users and other non-developers to continuously deploy their new and evolving knowledge as reusable software system components.

2. BACKGROUND
This section sets the socio-technical context for our research.

2.1 Industrial Analytics Ecosystems
Even in this digital age, many industrial enterprises rely intensively on physical assets, and so are seeking ways to make better-informed asset decisions. These decisions can have significant bottom-line impact on profitability, compliance, customer satisfaction, and other critical aspects of mission performance. As industrial devices and sensors become smarter and more interconnected, more opportunities arise to leverage IIoT data for decision support. For asset health solutions and for other industrial applications (such as intelligent alarm management), ABB has increasingly pursued blending business intelligence with advanced "industrial analytics" that incorporate deep knowledge of ABB's technical SMEs (Subject Matter Experts) on industrial equipment and verticals. In such IIoT ecosystems, professional software engineers generally architect the extensible applications and "software product lines", and still develop most solution components. Technical SMEs focus on creating "black box" data analysis plugins that incorporate their industrial knowledge. Data inputs and outputs are exposed, but valuable expertise inside the black box is protected. To simplify integration (and reduce cognitive burden on SMEs), these algorithms can be treated as stateless. These industrial analytics plugins form a smaller problem space with respect to end-user development and continuous deployment.

2.2 Technical SMEs as End User Developers

Technical SMEs are professional experts who spend their workdays deepening their knowledge of industrial equipment and systems which are the core focus of their jobs. Across various industrial domains, SMEs tend to use a diversity of tools to help them analyze data pertinent to the equipment or systems they know. Many SMEs are proficient to varying degrees with tools such as spreadsheets and databases. However, most SMEs do not consider themselves programmers or have interest in “becoming programmers” – writing software or analyzing data is “job 2” [9]. They simply want to do their jobs more effectively and efficiently, and to easily apply their knowledge in multiple software systems. The technical SMEs who provide the analytic algorithms for IIoT are clearly “end-user developers”, i.e. people who create software but whose primary job and expertise is not software development.

2.3 SME Roles in Deploying Analytics

Technical SMEs are not the only SMEs who participate in developing and delivering industrial analytics. In starting our research, the three SME roles listed in Table 1 were identified as crucial [11]. To comprehensively meet the goal of accelerating delivery time from idea to deployment, a supportive toolset must address the needs of all three types of SMEs.

Table 1. Key subject matter expert (SME) roles

Role	Key Competency
Technical SME	ABB and customer industrial equipment, operational systems, and domains
Solution SME	ABB and customer software systems and how expert analytic algorithms can enhance them
Integration SME	Deploying and tuning solutions and algorithms in customer operating environments

2.4 Challenges and Past Tactics

As end user developers, SMEs face substantial challenges in efficiently converting their knowledge and/or the software they’ve created as end users into executable software modules that can be reused in industrial software solutions. One common workaround involved close partnerships with software developers. Technical SMEs might generate non-executable documentation of their data analysis logic and hand off these requirements inputs to developer partners. Iterations of development and validation followed. Similar collaboration cycles were required for Solution SMEs to integrate these algorithms into various software solutions, and for Integration SMEs to tailor them for diverse deployment scenarios.

Although this human process has worked ‘in the small’, it has been slow and inefficient for all three types of SMEs, particularly in the long tail of maintenance and evolution. This approach is not scalable for the vast numbers and types of industrial equipment and systems in a comprehensive IIoT analytics solution, or for delivering the desired agility and speed to market.

3. RELATED WORK

In this section, we summarize work related to rapid software delivery, tool support for end-user developers, and deployment.

3.1 Rapid Software Evolution and Delivery

Many approaches have been pursued towards accelerating software system development and managing risks associated with integration and evolution, including model-based, aspect-oriented, and component-based development; agile methodologies; software product line architectures; and engineering systems of systems. In pursuit of continuous software evolution and delivery, a “stairway to heaven” has been described with five levels and four transitional stair steps (traditional R&D, agile R&D, continuous integration, continuous deployment, and continuous experimentation & learning from real-time customer usage) [5]. Building blocks for a continuous experimentation system and infrastructure have been presented [2], and the term “Continuous*” has been proposed for the holistic landscape of continuous initiatives [3]. To date, possible roles of end user developers in Continuous* have not received much attention.

3.2 Tools for End-User Development

Conventional programming languages (e.g., C# .NET, Java) and scripting languages (e.g., JavaScript, Python) are less commonly used by end user developers. To better support non-developers and novice programmers as well as professional software developers, significant research activity at universities has focused on “natural programming” toolsets and strategies (e.g. [7]) and end-user software engineering (e.g. [1] and [9]). Toolset vendors are also actively addressing end user needs. Macro-programming tools (e.g. MatLab or GNU Octave) can help non-developer SMEs with abstractions to specify application logic while hiding low-level details. Visual tools that require little or no coding can also be appealing for visual thinkers or those with limited programming expertise; examples include RapidMiner and SAS (Statistical Analysis System). Related tools offering drag-and-drop interfaces for quick prototyping include Node-RED for wiring together hardware devices and online services, and Touch Develop for quick creation of a mobile prototype.

In the industrial automation world, the Distributed Control System (DCS) represents a good example of a programmable system for SMEs. A DCS consists of function blocks which encapsulate coded process control algorithms, wired together by a SME who understands how to control the process, but who does not need to be a professional software developer. SMEs are insulated from the internal coding of function blocks. DCS toolsets generally have platform-specific internal representations of logic and are tightly coupled to built-in, platform-specific deployment mechanisms. For future, standard IEC 61499 offers the possibility that DCS vendors could offer platform-independent function blocks.

3.3 Algorithm Deployment

To deploy application modules (“apps”) on mobile devices to a large global audience, platform-specific marketplaces have been established, e.g. Google Play for Android and the Apple App store for iOS. Similarly, marketplaces are integral elements of the ecosystems for common Integrated Development Environments (IDEs) which enable others to extend their capabilities. For instance, the Eclipse Marketplace lists more than 1000 third-party plugins that are available for developers to use, and more than 48,000 extensions for Visual Studio are available in Microsoft’s NuGet Gallery. Our research is similar in spirit but targets reusable analytic applications and algorithms by SMEs, not IDE extensions or standalone mobile apps by developers.

4. TECHNICAL APPROACH

The research initiative began in 2012 [10] and is scheduled to execute through 2016. This section briefly describes the initial and evolved technical strategy we have pursued with respect to end user SME development and deployment of analytic models.

(For a more architectural and detailed view of results and lessons learned by early 2015, see [11]; herein, we focus mostly on the subsequent activities and lessons of the past 12 months.)

4.1 Goal: Continuous* Support for SMEs

The challenges summarized in sections 1 and 2 drove us to explore toolsets that could help to accelerate development cycles for SMEs to continuously deliver expert algorithms. The vision for SME Workbench is not to supplant existing analytics toolsets which SMEs may naturally find suited for their work, but to complement them. A key objective is to ease orchestration and reuse of SME models (whether built in-Workbench or in third party tools) as first-class elements of many software ecosystems.

4.2 Strategy: Toolset Product Line

To achieve the desired flexibility and multi-platform interoperability, the SME Workbench was designed as a runtime-extensible application suite with a cloud-based model store, following “dynamic software product line” (DSPL) architecture concepts. Figure 1 shows the Applications, Core, and Extensions in the SME Workbench architecture as it had evolved by mid-2015. *(Reference [10] provides an in-depth view of the technical strategy and product line; we summarize here only the key aspects as they relate to acceleration of delivery and the SME roles.)*

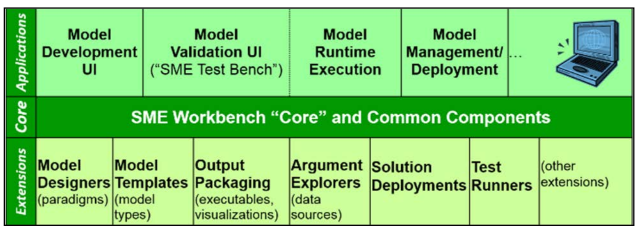

Figure 1. SME Workbench DSPL Components, 2015 © ABB

From this toolset product line, many SME Workbench “variants” can be assembled for internal or external use. Each variant has the Core plus one or more Applications and selected solution-relevant Extensions. For instance, with a variant for external use, an end customer’s technical SMEs could supplement ABB’s knowledge with their own, and deploy their analytic algorithms into a running industrial asset health solution. This further increases end-user content and dynamic variability of the asset health product line instance. This scenario magnifies the need for toolset mechanisms to support validation and to protect integrity and performance of the blended industrial analytics solutions.

4.3 2015-2016 Evolutions in SME Support

To support Solution SMEs, in 2015 we prototyped Extensions to auto-generate SME models in the exact executable formats their solutions need. These Extensions now include auto-generation of KPI-ML (XML) files to describe in a standards-based, machine-readable way how solutions can invoke each SME model.

We also prototyped new Deployment Extensions for solution platforms where it was desirable to enable technical SMEs to directly deploy their algorithms into a running instance without a solution SME or integration SME in the loop.

This 2015 work demonstrated feasibility and also confirmed the importance of the toolset supporting all SMEs in validating not only the functionality, but also the performance qualities of the new components.

The Model Development app now provides simple feedback on speed of execution. The Model Validation (“Test Bench”) app in the toolset is being prototyped in 2016, to not only enable bulk execution of models for validation, but also to provide SMEs with capabilities for simulation and property-based unit testing.

4.4 Domain-Relevant Modeling Paradigms

Many language paradigms for end-user development are possible, e.g. “spreadsheets, web services, MatLab, screen transitions, scientific, database, grid, control flow, dataflow, event” [1]. To align with the control flow paradigm we observed that technical SMEs had used to document their algorithmic ideas for software developer partners, we based the first Model Designer extension on Windows Workflow. A key design principle that gained increased importance in 2015 was *flexibility* to enable use or reuse of analytic algorithms from toolsets that SMEs may naturally prefer. For instance, early input from technical SMEs indicated that some liked to work in MatLab. Other SMEs who had already made substantial investments in conventional partnerships with software developers were seeking ways to reuse existing .NET DLL models. Accordingly, during 2015 we prototyped custom toolbox “activities” to ease reuse of existing SME models in MatLab or .NET DLL formats, and validated them with SMEs.

A key aspect of the 2015 work was engaging internal SMEs in pilots. Our 2015 pilot experiences affirm for end-user IIoT analytics the maxim observed 20 years ago: “*When seeking information, there must be a cognitive fit between the mental representations and the external representation.* If your mental representation is in control flow form, you will find a dataflow language hard to use; if you think iteratively, recursion will be hard.” [4] The control flow paradigm was well received by the SMEs selected to pilot it in mid-2015. As awareness of the toolset prototype spread within the company, new potential users emerged who were more comfortable with dataflow-based mental representations and domain-specific toolsets, rather than logic or control flow. We received quick feedback asking if we could create a new model designer extension that supported dataflow, and expressing that the toolset would be a “home run” for them if it just had that additional capability. The 2016 research plan now includes a proposed activity for extending the visual Model Designer prototype to provide dataflow elements in the toolbox.

4.5 Overall 2015 Pilot Results

Our pilot affirmed that the extensible toolset approach appears to be a great fit for efficiently supporting SMEs with the diversity of industrial equipment, domains, applications, and software solutions in IIoT ecosystems. Technical SMEs value being able to reuse advanced analytic models from familiar toolsets and paradigms. For simpler orchestrations or calculations, end-user SMEs appreciate the visual workflow designer and the self-documenting models they can create. An exemplary SME comment is *“I can’t do Visual Basic, but I can do this!”*

Using templates and code generation techniques provides a well-controlled environment that constrains the integration space for the benefit of Solution and Integration SMEs. Automated end-to-end deployment and integration support were also successfully piloted, and has provided the final piece to shrink the idea-to-deployment time from months to minutes.

We posit that toolsets enabling SME analytics can enhance agility and accelerate continuous deployments of evolving ideas and knowledge, via architectural support for controlled creation and delivery of product line components by SMEs. However, challenges remain in fully achieving our goals of agility, time to market, protection of qualities, and usability for SMEs.

5. OPEN ISSUES

This section briefly summarizes a few key areas related to our remaining challenges in supporting rapid end-user SME development and deployment of IIoT analytic solution plugins.

5.1 Role of End-Users in CSED

In a solution architecture empowering extensibility via end-user development, a typical deployed solution will blend controlled or provided content vs end-user content (e.g., perhaps 80% from solution provider/20% from end-users). Runtime extensibility with end-user components can clearly increase ecosystem agility by shrinking the time from idea to implementation to deployment, and accelerating feedback loops. However, preserving integrity and validity of industrial operations software requires extra care whenever end users are empowered to extend its functionality at runtime and potentially impact its critical performance qualities.

IIoT analytics carry certain further concerns and tradeoffs. For instance, protection of knowledge IP is critical to SME participation, but traceability and verifiability are key to customer reliance on black box algorithms. These considerations will escalate as more verification support is built into the toolset.

Improving ecosystem agility remains an important goal. We are exploring ways to enhance the value of the debugging and testing tools for all SMEs. To this end, and to both magnify the value and mitigate possible unintended adverse impacts of SME contributions, we see Continuous* tactics as adding value, and we are strengthening SME support in our toolset prototype. However, enablement of end-user development is known to present special benefits and challenges, and Continuous* for end user development is not yet a well-understood hybrid scenario.

We look forward to active discussions in the CSED'16 workshop on how toolsets for SMEs doing end-user development could better support secure, verifiable, rapid-to-continuous deliveries of industrial analytics within dynamically extensible ecosystems.

5.2 Measuring End User Toolset Usability

In our pursuit of Continuous* toolset support for SMEs, a key challenge is how best to measure usability and usefulness of various model design and integration tactics for end users.

For instance, we have not yet identified a satisfactory and efficient approach for comparing and contrasting suitability of various analytic model design tools and paradigms for a well-selected set of SMEs and models of varying complexities. Key elements include characterizing the SME's cognitive framework and technical background; measuring complexity of the analytic model the SME intends to build; and measuring usability to this SME of various toolsets for model building.

Measuring complexity of an algorithm idea in someone's head is clearly a challenge. Measurements of usability need to consider learning curves for multiple tools, as well as a per-model 'learning effect': the first implementation of a given model in any tool will likely be the slowest. These concerns might be mitigated by a careful plan for measuring usability and post-implementation complexity across multiple SMEs, models, and toolsets.

Similar challenges exist in evaluating debugging and integration features of end user toolsets. A proper study would need to consider validity challenges, e.g. as identified in [6] for tool usability studies, as well as challenges specific to measuring IIoT algorithm complexity and usability for technical SMEs with varying cognitive frameworks in diverse domains.

We welcome ideas and collaborators from the CSED community on ways to effectively measure and compare usability and benefit of toolsets supporting "Continuous*" for end user developers and for diverse types and complexities of IIoT analytic models.

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